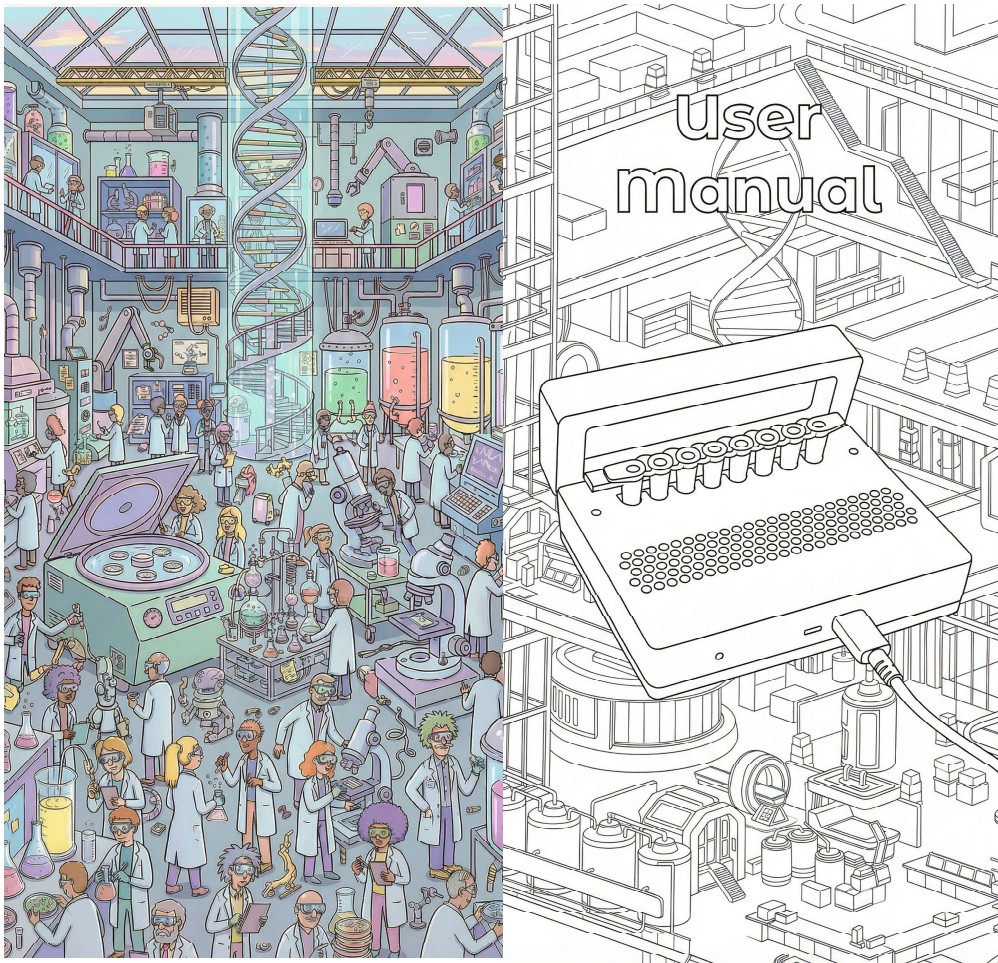




Repository



Preprint



First steps

1 WiFi

The **qByte connection** works through the local network. The hotspot does not need internet access; it only requires that your control device (i.e laptop) is connected to the same network.

1

When powered on, the qByte will try to connect to a default local network. If it cannot do so, it will create a temporary hotspot for configuration.



qByte-setup

Connect

Manage Wi-Fi settings

Connecting to qByte.local

2 Visit qByte.local

Once connected, open your web browser and visit the page **qByte.local**. You will see the temporary UI where you can configure a new local network and a mDNS. Enter your details and click on **Connect**. The qByte will restart and try to connect to the new network.



3 Revisit qByte.local

Connect your computer to the same local network, visit the page **qByte.local**, to access the control UI.



4 Final Tips

- 1** Unless you add oil over the reactions, **always** adjust the lid temperature to **20 °C** above the incubation temperature (e.g. 85 °C for 65 °C incubations).
- 2** If you perform long runs (e.g. melting ramps of 90–95 °C), you may obtain **better results without the lid** and with a layer of mineral oil.
- 3** If you use reactions with less than 15 µL, **first run a test reaction**. Do not assume that the optical response will behave as expected, as we haven't calibrated below that reaction volume.
- 4** This section will be expanded over time. If you encounter any issues, please **contact us** and we will include them.

✦ Troubleshooting ✦



Hotspot connection issues

Some hotspots are more difficult to connect than others. If you are using a **phone hotspot**, ensure the following:



✦ **Wi-Fi is disabled.** If it was enabled, turn it off and **restart** the hotspot.

✦ **Compatibility mode** is enabled (if available).



✦ **2.4 GHz** is selected as the hotspot frequency. Check this in the hotspot preferences.



mDNS conflicts (qByte.local)

If two devices have the **same mDNS name** (i.e. **qByte.local**), the last one to connect is the one that you access when typing **mDNS.local**.



mDNS not resolving correctly

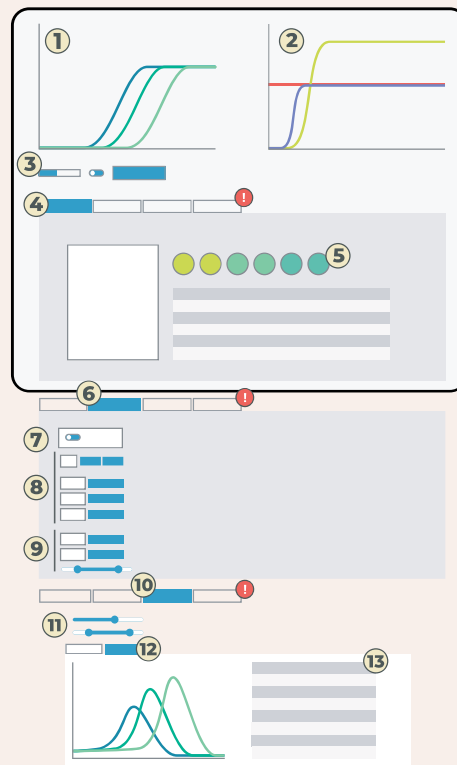
Sometimes devices do not update **mDNS** entries **immediately**. Laptops or phones may take time to refresh their local address table, or the hotspot may need to be **restarted**.

- Wait a few seconds and try again.
- **Restart** the hotspot.
- Disconnect and reconnect the device to the network.



3 User Interface

- 1 Raw fluorescence data
- 2 Raw temperature data
- 3 General controls
 - Available memory
 - Simply graph for performance
- 4 Protocol design tab
- 5 Tube selection
- 6 Experiment design tab
- 7 Mode (Incubation/Melting)
- 8 General settings
- 9 Melting settings
- 10 Analysis tab
 - 11 Range selection
 - 12 Normalization mode selection
 - 13 Result analysis
 - Time-to-threshold (Ct)
 - Click on the graph to set threshold
 - Melting peak



! The last tab is "Calibration", which is currently under development. It is intended for device manufacturers to adjust the temperature and fluorescence sensors before distribution.